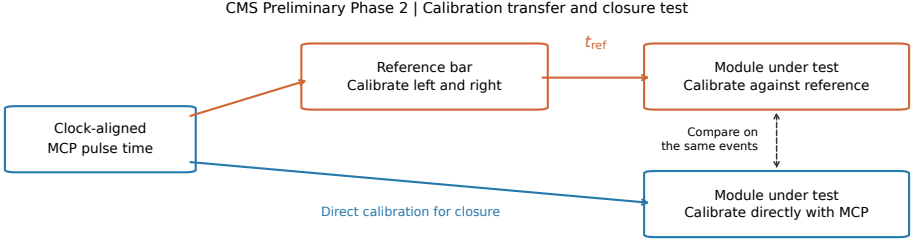
Transfer the MCP calibration to the module under test

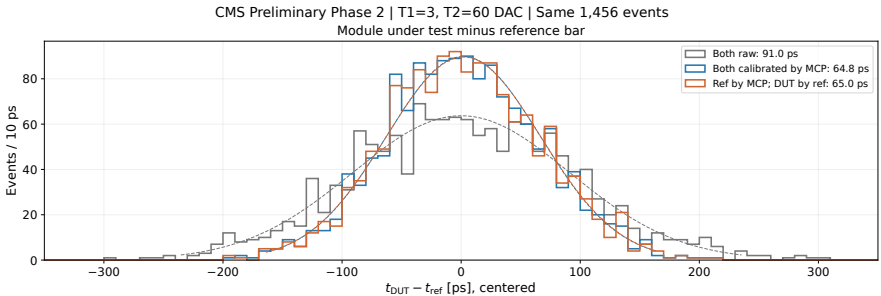
Purpose: use the calibrated reference bar when no MCP event time is available.



First form $t_{\mathrm{ref}} = \frac{1}{2}[(t_L - W_L^{\mathrm{MCP}}) + (t_R - W_R^{\mathrm{MCP}})]$. Then fit $t_j - t_{\mathrm{ref}}$ versus $1/Q_j$ for each end j of the module under test; subtract the resulting $W_j^{\mathrm{ref}}(Q_j)$.

The downstream transfer fit uses TOFHIR times and the saved reference-bar calibration, not the event's MCP time. For closure, also fit the downstream channels directly against MCP on the same selected sample. Both routes use 20 inverse-charge bins and SEM-weighted quadratics. One setting only: $T_1 = 3$, $T_2 = 60$ DAC. This tests transfer to one bar, not yet to every module or operating condition.

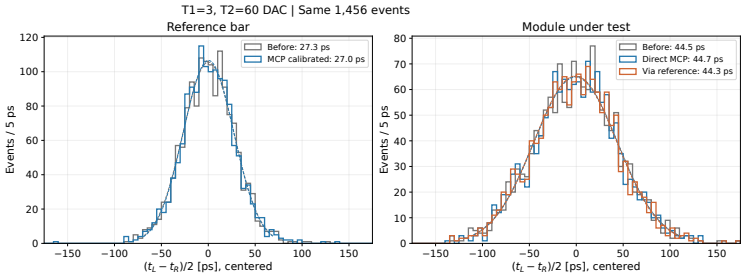
The transferred calibration agrees with direct MCP calibration



Both routes give a bar-pair width of about 65 ps, compared with 91.0 ps before correction.

Same 1456 events; common fitted charge domains, no extrapolation or threshold mixing. Bar times average the two corrected ends. Dashed curves: truncated Gaussian core fits within median $\pm 2.5(1.4826\,\mathrm{MAD})$; global median removed only for display. Width difference (via reference minus direct): $+0.24$ ps; 68% interval $[-0.30, 0.44]$ ps from 400 paired subset-bootstrap replicas. Cuts and calibrations fixed; no calibration or systematic uncertainty included. In-sample closure, not independent validation.

Backup: left-minus-right timing check



For $t_{\mathrm{bar}} = (t_L + t_R)/2$, estimate $\sigma_{\mathrm{bar}}^{(L-R)} = \sigma(t_L - t_R)/2$. Combine the bar estimates: $\sqrt{27.0^2 + 44.3^2} = 51.9$ ps, compared with the measured 65.0 ps bar-pair width.

The plotted variable is $(t_L - t_R)/2$: widths, fit curves and bins are rescaled by one half, with no refit. Same events as the closure test. Interpreting these widths as bar-average resolutions requires independent end fluctuations and negligible position spread; quadrature also assumes independent bars. Common timing fluctuations cancel in the half-difference. Correlations and position effects require further study to explain the 65 ps width.